Practice Set 4 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Discuss management of ureteric injuries occurring during gynec surgeries

Type: LAQ (25 marks) (also answered in Practice Set 2 Paper 3, Q1)

Introduction

- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit. The **distal ureter** is the usual site.
- The core problem is **recognition**: ligation, thermal and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

Incidence (EAU (European Association of Urology) trauma guideline data)

Procedure	Ureteric injury (%)
Caesarean section (emergency)	0.01–0.06
Vaginal hysterectomy	0.02–0.5
Abdominal hysterectomy	0.03–2.0
Minimally invasive hysterectomy	0.13–6.0 (laparoscopic = robotic, about 0.9)
Prolapse surgery	**1.7–4.3 (highest)**

- **Risk factors:** endometriosis (encasement, adhesions, medial deviation), placenta accreta, malignancy, radiotherapy, previous surgery, heavy bleeding, learning curve.

Applied anatomy and mechanisms

- **Danger sites:** pelvic brim at the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, transection, thermal injury, **devascularisation**. In about 10% of women the uterine artery supplies much of the distal ureter, so some ischaemic injuries are unavoidable.

Classification: AAST (American Association for the Surgery of Trauma)

Grade	Injury
I	Contusion or haematoma
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Transection, <2 cm devascularised
V	Avulsion, >2 cm devascularised

Prevention

- Identify the ureter at the pelvic brim (crossing the common iliac artery) and trace it before ligating pedicles; good haemostasis.
- **Prophylactic stents** (or lighted stents) help recognition in complex cases but do not clearly prevent injury. **Intraureteric ICG (indocyanine green)** fluoresces for hours.

- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.

Recognition at surgery (best outcome)

- Inspect the ureter; IV indigo carmine or methylene blue; cystoscopy for jets; retrograde pyelogram or stent; check the other ureter and bladder.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

Delayed presentation and diagnosis

- Mean diagnosis about day 10: fever, ileus, flank pain, rising drain output, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF) within 4 weeks.
- **UVF:** normal voiding plus constant leak, because the other kidney fills the bladder.
- **CT (computed tomography) urography with 10-minute delays** (both ureters). **Retrograde pyelogram** is most sensitive and allows stenting. Drain or vaginal fluid creatinine above serum confirms urine.
- **Double-dye test:** oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF (vesicovaginal fistula); both = both.

Management

Principles: debride to bleeding edges; spatulated, tension-free, watertight, stented repair with fine absorbable suture; preserve adventitia; drain; omentum for thermal injury; non-tunnelled reimplant.

At surgery, by injury

Injury	Action
Suture through wall, minor thermal	Remove suture; stent
Partial tear	Transverse closure (4-0/5-0) over stent, 4–8 weeks
Ligation	Deligate; stent if viable, excise and repair if not
Transection, lower ureter (usual)	**Ureteroneocystostomy ± psoas hitch** (95–100% success)
Long distal gap	**Boari flap** + hitch (gaps up to 12 cm)
Transection above pelvic brim	Ureteroureterostomy
Unstable patient	Ligate + later nephrostomy, or exteriorised stent

Delayed recognition

```
Suspected late injury
 -> retrograde stent (or nephrostomy + antegrade stent)
     |-- succeeds -> stent 4-8 weeks, then image
     |-- fails -> nephrostomy
          |-- found <=7 days -> early repair
          |-- later -> repair after about 6 weeks
Sepsis / uncontrolled collections -> drain; repair regardless of timing
```

- **UVF:** stent first, ideally within 2 weeks (most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic repair matches open; avoid nephrectomy.

Follow-up and medicolegal

- Ultrasound or diuretic renogram at 4–12 weeks and 1 year; strictures may appear up to 30 months.
- Disclose and document; malpractice claims most often follow gynaecological surgery.

Diagram to draw: ureter passing under the uterine artery; psoas hitch with non-tunnelled reimplant.

> **Key points to remember:**
>
> - Distal ureter under the uterine artery is the classic site; most injuries present late.
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; repair ≤7 days or after 6 weeks.
> - Psoas hitch 95–100%; Boari flap bridges up to 12 cm.

Q2. Evaluation and treatment of newborn with ambiguous genitalia

Type: LAQ (25 marks) (DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

Introduction

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours.

Classification (by karyotype)

Group	Main causes
46,XX DSD	**CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive)
46,XY DSD	Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); PAIS (partial androgen insensitivity); LH (luteinising hormone) receptor defect
Sex chromosome DSD	45,X/46,XY MGD (mixed gonadal dysgenesis); chimerism

History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven otherwise. Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

Investigations

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (earlier values are falsely high); electrolytes and glucose daily. **Salt-wasting appears around day 5–7**, so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

Test	Meaning
AMH (anti-Müllerian hormone)	Present = functioning Sertoli cells

Test	Meaning
Testosterone, LH, FSH (follicle-stimulating hormone)	Testicular function (mini-puberty)
Androstenedione, DHEA-S (dehydroepiandrosterone sulfate)	Adrenal enzyme block
Testosterone:DHT (dihydrotestosterone) >20:1	5α-reductase deficiency
hCG (human chorionic gonadotropin) stimulation	No rise = no Leydig tissue or LH receptor defect

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH).
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels, microarray, exome. **No gonadectomy during diagnosis.**

Diagnostic algorithm

```
Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular;
                 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect)
46,XX + uterus + high 17-OHP           -> CAH
46,XY, no uterus                       -> androgen synthesis/action defect
46,XY or 45,X/46,XY + uterus           -> gonadal dysgenesis / MGD
```

- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

Treatment

1. Emergency (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.

2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

3. Sex of rearing

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

4. Surgery: timing and consent

- Irreversible surgery is ideally deferred until the child can participate; parents decide with the team after balanced counselling.
- **Feminising:** clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often in infancy (complication rates high).

5. Gonads

Risk	Condition	Plan
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	Keep CAIS gonads until after puberty

Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

Diagram to draw: Prader stages I–V and the urogenital sinus with confluence level.

> **Key points to remember:**
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty.

Q3a. Botox in uro

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 2, Q3d)

Read as: botulinum toxin in urology.

Introduction and mechanism

- *Clostridium botulinum* neurotoxin; urology uses **onabotulinumtoxinA (BoNT-A)**: heavy chain (binding, entry) plus zinc-dependent light chain.
- Light chain **cleaves SNAP-25** (synaptosomal-associated protein 25), so vesicles cannot fuse and **acetylcholine release stops**.
- **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide); fewer TRPV1 and P2X3 receptors, so urgency falls.
- Effect wanes as nerve terminals sprout (6–9 months); repeat injections work.

Technique

- Cystoscopic detrusor injection: **100 U at 20 sites or 200 U at 30 sites**, trigone usually spared; check PVR (post-void residual) at 2 weeks.

Uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK). ABC trial: equal to antimuscarinic, more complete dryness. ROSETTA: 200 U slightly better than SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY): less incontinence, lower detrusor pressure
Paediatric NDO	FDA (Food and Drug Administration) label 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U (often trigonal); AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	Intraprostatic: large RCTs (randomised controlled trials) no better than placebo; not recommended

- **Also know:** pelvic floor spasm; chronic pelvic pain. Units of different toxin products are not interchangeable.

Complications and contraindications

- **Retention needing CISC (clean intermittent self-catheterisation)**, commoner in NDO; each 10 mL of baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- UTI, haematuria; rarely generalised weakness.
- Contraindications: active UTI, myasthenia gravis, aminoglycosides, pregnancy.

Diagram to draw: cholinergic nerve terminal showing SNAP-25 cleavage.

Key points to remember:

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO.
- Main risks: retention needing CISC, UTI.

Q3b. Priapism

Type: SAQ (10 marks) (full LAQ answer in 2025-Winter Paper 3, Q2)

Definition and types

- Erection lasting **≥4 hours** beyond, or unrelated to, sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2021–2022).

	Ischaemic (95%)	Stuttering	Non-ischaemic (high flow)
Lesion	Veno-occlusion	Recurrent ischaemic episodes	Arterio-sinusoidal fistula
Penis	Rigid, painful	Recurrent, often nocturnal	Partial, painless
Cause	ICI (intracavernosal injection), drugs, SCD (sickle cell disease)	SCD	Perineal trauma
Urgency	**Emergency**	Per episode	Not urgent

Pathophysiology

- Ischaemic: compartment syndrome; hypoxia and acidosis cause thrombosis, smooth-muscle necrosis and fibrosis, hence ED (erectile dysfunction).
- Stuttering: low basal NO (nitric oxide) downregulates PDE5 (phosphodiesterase 5), so cGMP rises unopposed.

Diagnosis

- **Corporal blood gas:** ischaemic **pO2 <30, pCO2 >60 mmHg, pH <7.25**; high flow resembles arterial blood (pO2 >90).
- **Penile duplex:** absent cavernosal flow (ischaemic) or high flow with fistula.
- Blood count, haemoglobin electrophoresis, toxicology, without delaying treatment.

Management of ischaemic priapism

```
Dorsal block -> aspiration ± saline irrigation
 + phenylephrine 100-500 µg/mL, 1 mL every 3-5 min, up to 1 h
   (monitor blood pressure and pulse)
 -> fails -> distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling
 -> proximal shunt (Quackels) / penoscrotal decompression
Beyond 36 h or refractory -> early penile prosthesis
```

- **ED risk by duration (SCD series):** ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- Early prosthesis: less shortening and fewer revisions than delayed insertion.

Stuttering

- Home phenylephrine; prophylaxis with daily PDE5 inhibitor, pseudoephedrine, etilefrine; **also know:** antiandrogens, GnRH (gonadotropin-releasing hormone) agonists, baclofen.

Non-ischaemic

- Observe with compression and ice (about 60% resolve); then **selective arterial embolisation** (resolution ≥75%); surgical ligation last.

Complications

- ED, corporal fibrosis, penile shortening; shunt-related urethral injury, fistula, infection and skin necrosis.

Diagram to draw: distal and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type.
> - Aspiration plus phenylephrine first, then distal shunt.
> - ED 100% beyond 36 h; consider early prosthesis.

Q3c. Oligospermia

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 2, Q3a)

Definition

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO (World Health Organization) 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million). The legacy cut-off was 20 million/mL.
- These centiles come from fertile men; they are a reference, not a fertility test. Define oligospermia in context (testis size, FSH (follicle-stimulating hormone), repeated values).
- **Severe: <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

Other WHO 2021 5th centiles	Value
Volume	1.4 mL
Progressive motility	30%
Normal forms (strict)	4%
Vitality	54%

Causes

- **Pre-testicular:** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, hyperprolactinaemia, obesity (raised oestradiol), opioids.
- **Testicular: varicocele** (commonest correctable; heat and oxidative stress), cryptorchidism, mumps orchitis, torsion, gonadotoxins, Klinefelter mosaic, **AZFc (azoospermia factor c) deletion**; idiopathic (largest group).
- **Post-testicular:** partial EDO (ejaculatory duct obstruction; low volume), infection, antisperm antibodies.

Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days; parameters peak at 1–2 days).
- History, testis volume, varicocele, vasa.
- Morning testosterone, LH (luteinising hormone), FSH; oestradiol and prolactin if testosterone low.
- **<5 million/mL: karyotype and Y-microdeletion** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)).
- Selected: TRUS (transrectal ultrasound) if volume <1.0 mL; DNA fragmentation; semen culture.

Management

- Stop testosterone and gonadotoxins; weight loss; stop smoking.
- **Microsurgical varicocelectomy** for a **palpable** varicocele with abnormal semen; never for subclinical.
- **Medical:** hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH. Idiopathic with low testosterone: clomiphene 25 mg daily (off-label); anastrozole 1 mg daily if testosterone:oestradiol <10.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc deletion.
- Spermatogenesis takes about 64 days; reassess at 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with treatment sites.

Key points to remember:

- WHO 2021: 16 million/mL; severe <5 million/mL.
- Karyotype and Y-microdeletion if <5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

Q3d. Immunobead test for semen

Type: SAQ (10 marks)

Introduction

- The IBT (immunobead test) detects **antisperm antibodies (ASA)** bound to sperm (direct) or free in body fluids (indirect).
- **Pathogenesis:** Sertoli tight junctions form the blood–testis barrier. If it is breached, sperm antigens meet the immune system. ASA cause **agglutination, poor motility, poor cervical-mucus penetration and impaired fertilisation**.
- **Causes:** vasectomy and other obstruction; orchitis, prostatitis, STI (sexually transmitted infection); varicocele, cryptorchidism, heat; trauma, torsion; genetic (HLA-B28).

Indications

- Sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

Principle and method

- Polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

Test	Method	Use
Direct IBT	Patient's motile sperm washed free of seminal plasma; mixed with beads; motile sperm with attached beads counted	Sperm-bound antibody, class and site
Indirect IBT	Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done	Free antibody; patient sperm too few or immotile; female partner
MAR (mixed antiglobulin reaction)	Fresh unwashed semen + IgG (or IgA)-coated latex beads + anti-IgG; mixed sperm–bead clumps counted	Quick, cheap screen; IgG/IgA only

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm. **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

Management

- Condoms and sperm washing lack good evidence (washing removes unbound, not bound, antibody).
- Corticosteroids: two controlled trials conflict.
- IUI for mild binding; **IVF with ICSI (intracytoplasmic sperm injection)** is the usual treatment, bypassing antibody effects.

Diagram to draw: motile sperm with beads attached at head and tail.

Key points to remember:

- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Treatment of choice: ICSI.

Q3e. Duloxetin

Type: SAQ (10 marks)

Read as: duloxetine, mainly its role in stress urinary incontinence.

Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

Diagram to draw: Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

Key points to remember:

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA.

Q3f. CBAVD

Type: SAQ (10 marks)

Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.
- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

Technique	Points
MESA (microsurgical epididymal sperm aspiration)	Best: many motile sperm; cryopreserve several vials
PESA (percutaneous epididymal aspiration)	Local anaesthetic; fewer sperm
TESA/TESE (testicular aspiration/extraction)	If epididymal retrieval fails

- ICSI results are good; one study reported more miscarriage and stillbirth.
- **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis.

Diagram to draw: absent vas and seminal vesicle with intact testis and caput epididymis.

Key points to remember:

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Rapid Revision Sheet – Practice Set 4 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss management of ureteric injuries occurring during gynec surgeries (LAQ)

- Distal ureter under the uterine artery is the classic site; most injuries present late.
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; repair ≤7 days or after 6 weeks.
- Psoas hitch 95–100%; Boari flap bridges up to 12 cm.

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- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
- 17-OHP after 36 h; salt crisis around day 5–7.
- Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
- Defer irreversible surgery; CAIS gonads kept until after puberty.

Q3a. Botox in uro (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO.
- Main risks: retention needing CISC, UTI.

Q3b. Priapism (SAQ)

- Blood gas decides the type.
- Aspiration plus phenylephrine first, then distal shunt.
- ED 100% beyond 36 h; consider early prosthesis.

Q3c. Oligospermia (SAQ)

- WHO 2021: 16 million/mL; severe <5 million/mL.
- Karyotype and Y-microdeletion if <5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

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- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
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Q3f. CBAVD (SAQ)

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.